

Versioned Publication Development Record

Dr. William James Stanley | 11 September 2026 | v0.4.0

Millennium Problems, AI, and Scientific Humility

Purpose

This file preserves the development of one argument across two publication layers. The LinkedIn version is intentionally narrow, immediate, and publicly legible. The website essay version retains the deeper philosophical and epistemic argument but remains deliberately unfinished so the author can continue shaping it.

No private system architecture, implementation details, research mechanics, or IP-sensitive methods are disclosed. The author's work with the Hodge conjecture is described only at the level required to explain the epistemic purpose of the exercise.

Version lineage

v0.1 - Original position / conceptual source

The originating position was that the objection is not to AI doing mathematics. The objection is to the posture of treating historically accumulated scientific problems primarily as capability demonstrations. The key propositions were:

- AI is a legitimate scientific tool and may become an extraordinary mathematical instrument.
- A major open problem is not an isolated puzzle; it is the exposed frontier of generations of definitions, lemmas, failed routes, representations, teaching, criticism, and partial progress.
- Scientific problems therefore have ancestry and contain people.
- Computational power does not remove scientific obligations such as provenance, attribution, criticism, uncertainty, patience, reproducibility, and intellectual generosity.
- The author's own use of the Hodge conjecture is different in purpose: the difficulty is used to expose the boundary of understanding rather than merely to obtain an impressive output.
- The broader philosophical concern is that modern technological culture can mistake the removal of friction for progress even when the difficult middle is where understanding is formed.

A central image emerged from the original formulation: the author is not trying to smash through a window because reaching the other side would look impressive; the author is trying to understand who built it, why it has its particular shape, what can legitimately be seen through it, and what remains unknown.

v0.2 - Structured long-form argument

The argument was reorganised into a more formal progression:

- 1 A scientific problem is not merely a problem.
- 2 The capability-test framing is understandable from a systems perspective but philosophically incomplete.
- 3 Scientific power does not cancel scientific obligation.
- 4 The author's own Hodge work is a disciplined test of understanding, not a claim to effortless conquest.
- 5 There is a difference between obtaining an answer and earning understanding.

6 Scientific humility should scale with scientific power.

This version established the distinction between **instrument** and **spectacle** as the central conceptual boundary.

v0.3 - Technical/philosophical amalgamation

The language was sharpened to signal familiarity with mathematics, philosophy of science, epistemology, provenance, benchmark design, and scientific norms without becoming jargon for its own sake.

The key formulation became:

The problem with using the Millennium Prize Problems as an AI benchmark is not the AI. It is the epistemology.

The author's Hodge work was described as **epistemic stress testing**: moving backwards through definitions, theorem dependencies, assumptions, representations, and missing prerequisite knowledge whenever reasoning fails. This remains the preferred high-level description because it explains the intellectual method without exposing private mechanisms or implementation details.

v0.4A - LinkedIn publication draft

Scientific Power Without Scientific Humility

The problem with using the Millennium Prize Problems as an AI benchmark is not the AI. It is the epistemology.

OpenAI's decision to direct large-scale model capability toward the remaining Millennium problems has bothered me more than I expected. Not because machine-assisted mathematics is illegitimate; quite the opposite. If AI can traverse proof spaces, generate conjectural structure, test formal consistency, or identify mathematical relationships at scales inaccessible to individual researchers, it could become an extraordinary scientific instrument.

What concerns me is the framing. A Millennium problem is not merely a difficult target in an abstract solution space. It is the exposed edge of a historical knowledge structure: definitions, partial results, failed proof strategies, counterexamples, new abstractions, changes in representation, and entire mathematical languages built over generations.

Science has ancestry. Hodge, Riemann and Navier-Stokes do not exist as isolated puzzles waiting for enough compute. Someone constructed the formalism. Someone proved the enabling lemma. Someone discovered that an attractive route was structurally impossible. Someone spent decades developing a framework that later became background notation. An open problem contains all of that, which means it contains people.

That is why I think the important distinction is not human versus machine. It is **instrument versus spectacle**. There is a substantial philosophical difference between asking whether a new system can make a legitimate contribution to an existing mathematical research programme and asking whether solving one of humanity's hardest problems demonstrates the power of the system. The computation may overlap; the scientific posture does not.

I say this while doing something superficially similar myself. I have been using the Hodge conjecture as a stress test for my own reasoning and systems work. I am not trying to brute-force a proof. I use a problem this deep because weak understanding becomes visible quickly: when the reasoning breaks, I have to move backwards through definitions, theorem dependencies, assumptions, representations, and the mathematics I simply do not know yet.

That is not answer extraction. It is epistemic stress testing. The problem is useful precisely because it exposes ignorance rather than allowing me to conceal it behind a plausible output.

AI may eventually contribute to a proof of one of these problems, and I hope it does. But participation in science comes with scientific obligations: provenance, attribution, reproducibility, criticism, uncertainty, and intellectual humility. Greater capability should increase those obligations, not dissolve them.

The frontier is not simply where the answer is missing. It is where accumulated human understanding currently stops. Nobody reaches that frontier alone - not mathematicians, not scientists, not models, and not the companies that build them.

Dr. William James Stanley

v0.4B - Website essay working draft

The Problem Is Not the AI. It Is the Epistemology.

OpenAI's decision to use the Millennium Prize Problems as a large-scale test of model capability has bothered me for a reason that initially seemed difficult to name. I do not object to AI doing mathematics. I use AI myself to study material far beyond my formal training, and I think sufficiently capable systems could become extraordinary scientific instruments. My objection is to something older than artificial intelligence: the relationship between intellectual power, inherited knowledge, and humility.

A great open problem is easy to misrepresent as an empty box with an answer missing from it. That picture is convenient for benchmarking because it gives us a clean before and after: unsolved, then solved. But the representation is wrong. Problems such as the Hodge conjecture, the Riemann hypothesis, and Navier-Stokes sit at the visible edge of enormous structures of prior work. Their statements depend on languages people had to invent, objects people had to define, theorems people had to prove, and failures people had to preserve so that others could move differently.

The frontier of mathematics is therefore not empty territory. It is densely inhabited by intellectual history. A proof, if one arrives, does not descend onto untouched ground. It lands on a structure built by generations of researchers, including people whose contribution may now survive only as a standard lemma, a notation convention, a discarded route, or a few lines in somebody else's paper.

This is why the idea of a Millennium problem as a capability benchmark makes me uneasy even though I understand the engineering logic behind it. If a system appears unusually strong at formal reasoning, one obvious way to probe its boundary is to direct it toward problems that have resisted exceptional human effort. That is not irrational. In fact, my own work contains a superficially similar move.

I have been using the Hodge conjecture as a stress test for my own reasoning and systems work. The details of that work remain private, but the epistemic principle is simple: use a problem difficult enough that superficial understanding fails quickly. When I cannot justify a step, I move backwards through its dependencies. What definition am I relying on? Which theorem permits the inference? What assumptions have entered unnoticed? Am I confusing a representation with the thing represented? What prerequisite mathematics am I missing altogether?

The aim is not to make the problem surrender. The aim is to make ignorance visible. In that sense, difficulty is not merely resistance; it is information.

That distinction leads to the philosophical boundary I care about most: **instrument versus spectacle**. A scientific instrument is used to extend inquiry while remaining accountable to the knowledge structure into which it enters. A spectacle uses the scientific object partly as evidence of the instrument's own power. Those activities can involve similar computations while embodying very different relationships to science.

If AI becomes capable of genuine mathematical discovery, the correct response is not to diminish its role. It is to take that role seriously. Participation in science should bring obligations with it: provenance, attribution, reproducibility, criticism, uncertainty, patience, and respect for prior work. A system may search a mathematical space at scales no individual person could match, but scale does not erase ancestry.

There is a larger argument here about education, technological friction, and the difference between obtaining outputs and developing understanding. I do not want to finish that argument yet. It deserves its own treatment rather than being compressed into an appendix to a dispute about AI benchmarks.

For now, the narrower principle is enough: scientific power and scientific maturity are different achievements. We should not assume that acquiring more of the first automatically gives us more of the second.

The frontier is not simply the place where nobody has written the final proof. It is the point at which a long human chain of definitions, failed attempts, criticism, teaching, abstraction, and discovery currently ends. Whatever reaches beyond it will still have arrived by standing on that history.

Provenance notes

- ****Human-origin position:**** The substantive philosophical position, the personal relationship to the Hodge conjecture, the distinction between learning from the problem and conquering it, the concern about scientific ancestry, and the broader criticism of technological shortcut culture originate with Dr. William James Stanley.
- ****AI transformation:**** GPT-5.6 Sol reorganised, compressed, and edited the supplied material into publication-oriented forms, introduced some connective terminology, and helped articulate the instrument-versus-spectacle distinction as a concise organising phrase.
- ****IP boundary:**** No private architecture, algorithms, implementation methods, system internals, or unpublished mathematical strategy are recorded here.
- ****Publication status:**** Neither publication version should be treated as final without the Human Gate. Specific factual descriptions of OpenAI's Millennium activity should be sourced and verified before external publication.
- ****Versioning:**** This record is append-oriented. Future revisions should add new numbered sections rather than silently replacing earlier states.